

CT071-3-3-DDAC - DESIGNING & DEVELOPING CLOUD APPLICATIONS**GROUP PROJECT**

Lecturer(s) :
Date Assigned : **Week 7**
Date Due : **Week 14**

LEARNING OUTCOMES

At the end of this project, you should be able to:

CLO1	Discuss the core concepts of the cloud computing paradigm and its future trends. (C2, PLO1)	Final Exam
CLO2	Design an application using recommended performance criteria that is suitable for deployment in a cloud platform. (C6, PLO2)	Assignment - Documentation
CLO3	Select the appropriate cloud services to build a high-availability and cost-effective application. (A3, PLO6)	Assignment - Implementation

PROBLEM BACKGROUND #1:

The rapid growth of internet usage has led to a significant increase in online scams that exploit individuals' trust, lack of awareness, and digital vulnerabilities. These scams range from phishing emails and fake job offers to fraudulent e-commerce sites and social media impersonation. Many users fall victim to such scams due to deceptive tactics that often appear legitimate. Scammers use psychological manipulation, urgent language, and fabricated identities to deceive individuals into revealing sensitive information or making unauthorized payments.

The issue is compounded by the lack of awareness and real-time information about ongoing scam trends. Victims often do not report incidents due to embarrassment, lack of knowledge on how to proceed, or fear of repercussions. Additionally, the widespread nature of scams across platforms such as email, social media, and messaging apps makes it difficult to track, analyse, or identify patterns. Many regions lack centralized systems that allow people to report scams easily or access verified information to protect themselves.

The absence of a common understanding of how online scams operate has created a gap in digital literacy, leaving vulnerable groups—such as the elderly, teenagers, and non-tech-savvy users—at greater risk. The evolving tactics of scammers make it hard for law enforcement and the public to keep up. This growing issue has not only led to financial losses but also emotional distress and mistrust in digital platforms.

Despite efforts from government and cybersecurity organizations, online scams continue to thrive due to inadequate public engagement and fragmented information. The problem highlights a need for a broader understanding of scam behaviour, public communication gaps, and the digital threats individuals face daily in an increasingly connected world.

PROBLEM BACKGROUND #2:

Diabetes is a chronic and complex disease affecting millions of people globally, requiring lifelong care and management. Individuals diagnosed with diabetes face daily challenges related to maintaining blood glucose levels, balancing food intake, exercising regularly, and adhering to prescribed medications. These challenges are often intensified by stress, lack of routine, and limited access to health education. Improper management can result in severe complications, including cardiovascular disease, kidney failure, nerve damage, and vision impairment. Many individuals, particularly those from low-income or rural backgrounds, may not have the necessary knowledge or resources to understand and manage their condition effectively. Language barriers, cultural perceptions, and a lack of consistent health guidance further contribute to the problem. Often, diabetes goes undiagnosed or untreated for long periods, leading to avoidable medical emergencies.

Moreover, patients often experience isolation and mental health issues as they navigate the day-to-day burdens of managing their illness. The continuous self-monitoring, strict dietary controls, and fear of complications can be overwhelming. Inconsistent communication between patients and healthcare providers adds to the difficulty, as does the absence of a personalized support system. The problem becomes more pronounced with the increasing prevalence of Type 2 diabetes linked to sedentary lifestyles, poor dietary habits, and limited awareness. Early intervention and prevention efforts are often undermined by lack of awareness campaigns or targeted health initiatives. Children and young adults are also increasingly being diagnosed, raising concerns about long-term health burdens on families and healthcare systems.

Despite the availability of treatment, the lack of an integrated approach to education, support, and health tracking leaves many patients struggling to manage their diabetes. This situation highlights the growing challenge of addressing chronic diseases in a rapidly changing global health landscape.

PROBLEM BACKGROUND #3:

Alzheimer's disease is a progressive neurodegenerative condition that affects memory, thinking, and behavior, primarily impacting older adults. One of the most distressing challenges associated with Alzheimer's is the tendency for individuals to wander and become disoriented, even in familiar surroundings. This wandering behavior can result in elderly individuals becoming lost, putting their safety at serious risk. They may not remember their name, address, or how to seek help, which complicates recovery efforts. Families and caregivers live in constant fear of their loved ones going missing and often lack the resources or tools to act swiftly when such situations arise. Missing elderly individuals may be exposed to various dangers including traffic accidents, falls, weather exposure, and even exploitation or abuse. The longer they remain unlocated, the higher the risk to their health and survival.

Efforts to locate missing seniors are frequently delayed due to the absence of timely information sharing, lack of centralized reporting, and minimal coordination between community members and authorities. While families may alert police or post on social media, these efforts are fragmented and often not seen by the right people in time. This creates a serious gap in the ability to quickly and effectively respond when a vulnerable individual goes missing. Furthermore, there is limited awareness among the public about how to identify and assist a person who appears lost due to Alzheimer's. Many people may overlook or misinterpret the signs, further delaying intervention. As aging populations continue to grow globally, the frequency of such incidents is expected to rise, placing additional pressure on families, caregivers, and communities.

This issue highlights the urgent need for improved awareness, communication, and responsiveness in dealing with missing elderly individuals suffering from Alzheimer's disease, a growing concern in both urban and rural settings.

PROBLEM BACKGROUND #4:

Flooding is one of the most destructive and recurring natural disasters, impacting millions of people every year. It causes loss of life, widespread property damage, disruption of infrastructure, and long-term economic setbacks. The unpredictability of weather patterns, especially with the increasing effects of climate change, has made it more difficult to anticipate when and where floods will occur. Many communities, especially those in low-lying or river-adjacent areas, remain vulnerable due to inadequate early warning systems and poor disaster preparedness. Accurate weather forecasting and flood prediction remain challenging in regions where infrastructure for data collection and monitoring is outdated or insufficient. Limited coordination between weather agencies and local authorities further complicates the situation, leading to delayed alerts and miscommunication. Residents are often uninformed about the severity of incoming storms or water level changes until it is too late to evacuate or take precautionary measures.

Vulnerable groups, including children, the elderly, and people with disabilities, face greater risks during floods, especially when emergency planning and communication are not inclusive. The lack of educational outreach and accessible flood-related information means that many individuals are unprepared to respond quickly in a crisis. Urban areas with poor drainage systems and rural regions with limited access to communication tools are particularly at risk. The damage from floods also has long-term impacts, including displacement, loss of income, and increased public health risks. The inability to predict floods accurately or communicate danger effectively continues to put lives at risk. This situation underlines the persistent gaps in awareness, access to timely information, and infrastructure needed to manage and reduce the risk of flooding, particularly in areas where such disasters are frequent.

QUESTION:

Your team is tasked with developing a cloud-based web application that addresses **one (1)** of the issues outlined above. The application must leverage cloud technologies to **ensure high performance and availability** by integrating ***various cloud services*** such as cloud computing, database, and storage solutions. ***You may use any programming language for development.*** Upon completion, the application must be deployed to the AWS cloud platform and utilize AWS monitoring tools to analyse its performance and operational metrics.

Rules of Project

1. Create a team comprising **THREE (3)** or **FOUR (4)** students.
2. Register your group in the Moodle system.
3. The **exact topic** must be agreed upon with the lecturer before your team start.
4. However, the actual content of the website is entirely up to your team and must be within the context of this module.
5. This group project will be divided into **2 different tasks** with overall **50 Marks**, which are:
 - Task #1 – Frontend & Backend Development with Cloud Compute and Database services (30 Marks)
 - Task #2 – Integrating Other Cloud Services and Monitoring Tool Components – 20 Marks.
6. This group project requires you and your team to share the workload equally.
7. Each student must **independently take on at least one distinct user role within the system**, with a minimum of two (2) unique system functionalities developed and implemented solely by them.
8. These functionalities must not overlap with those assigned to other team members.
9. A clear workload matrix must be provided, detailing the individual contributions of each student, which will be used for evaluation to ensure accountability and prevent duplication of work.

References

#CHILD: Acute need for autism services in Malaysia. (n.d.). Retrieved June 30, 2024, from <https://www.nst.com.my/lifestyle/heal/2024/04/1033304/child-acute-need-autism-services-malaysia>

#HEALTH: Depression has doubled in Malaysia. (n.d.). Retrieved June 30, 2024, from <https://www.nst.com.my/lifestyle/heal/2024/06/1065085/health-depression-has-doubled-malaysia>

Stem rising tide of youth violence. (n.d.). Retrieved June 30, 2024, from <https://thesun.my/opinion-news/stem-rising-tide-of-youth-violence-IF12424387>

PERFORMANCE CRITERIA

- Please refer to the Marking Scheme.

PLAGIARISM

Verbatim reproduction of material from external sources (web pages, books, papers, etc.) is not acceptable. If you are paraphrasing external content (or even your own prior work) or were otherwise influenced by them while completing your assignments, projects or exams, you must clearly acknowledge them. When in doubt, add a citation.

Grading:

Grade		Range
A+	Distinction	>=80%
A		75-79%
B+	Credit	70-74%
B		65-69%
C+	Pass	60-64%
C		55-59%
C-		50-54%
D	Marginal Fail	40-49%
F+	Fail	30-39%
F		20-29%
F		0-19%

APPENDIX 1

Marking Band		Band Description
Characteristics of Student Achievement by Marking Band	80-100 %	Exceptional problem solving with supporting analysis and application development skills with extraordinary originality and adoption of new tools, platforms, methods & technologies. Work may be considered for further expansion.
	75-79 %	Outstanding problem solving and application development skills.
	70-74 %	Excellent knowledge base that supports analysis, evaluation, problem-solving and application development of discipline with considerable originality
	65-69 %	Good knowledge base that supports analysis, evaluation, problem-solving and application development.
	60-64 %	Satisfactory knowledge base that supports some analysis, evaluation and problem-solving and application development.
	55-59 %	Basic knowledge base with some omissions at the level of technology. Restricted ability to discuss and/or or solve problems.
	50-54 %	Limited knowledge base. Limited understanding of discipline. Difficulty with problem solving and application development.
	40-49 %	Little evidence of knowledge base. Little evidence of understanding of discipline. Significant difficulty with problem solving and application development.
	30-39 %	Inadequate knowledge base. Inadequate understanding of discipline. Major difficulty with problem solving and application development.
	20-29 %	No evidence of knowledge base; no evidence of understanding of discipline. Total inability of problem solving and application development.
	0-19 %	Awarded for: (i) non-submission; (ii) dishonesty; (iii) in situations where the student fails to address the project brief (e.g.: answers the wrong question) and/or related learning outcomes